

What does `<!DOCTYPE html>` do?

HTML Foundations

It tells the browser to use modern HTML standards behavior for an HTML5 document.

```
<!DOCTYPE html>
```

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What is the root element of an HTML page?

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`<html>`. It contains all other document elements.

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How do you declare UTF-8 character encoding?

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Place `<meta charset="UTF-8">` inside `<head>`.

```
<meta charset="UTF-8">
```

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What is the largest HTML heading?

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`<h1>`, the highest-level heading.

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What is a void element?

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An element with no child content or closing tag, such as `<br>`, `<img>`, or `<source>`.

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Which image attribute supports basic accessibility?

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`alt`. Use descriptive text for informative images and `alt=""` for decorative images.

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How does a link open in a new tab?

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Use target="_blank" on the <a> element.

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Which element creates an unordered list?

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; each item is placed in .

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Which element defines a table row?

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`<tr>`. Header and data cells use `<th>` and `<td>`.

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Why use semantic HTML?

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It communicates the meaning and purpose of content to browsers and assistive technologies.

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What does a form's method attribute define?

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The HTTP method used to submit data, commonly `GET` or `POST`.

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How do you link an external CSS file?

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Use `<link rel="stylesheet" href="styles.css">` inside `<head>`.

```
<link rel="stylesheet" href="styles.css">
```

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Which input type creates a slider?

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`<input type="range">`.

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What does a form's action attribute define?

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The destination URL that receives the submitted form data.

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How is a label tied to an input?

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Match the label's for value to the input's id value.

```
<label for="email">Email</label>  
<input id="email">
```

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Which element embeds native video?

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<video>, often containing one or more <source> elements.

HTML Foundations

Which common element is inline by default?

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`<span>`.

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What is the syntax for an HTML comment?

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`<!-- This is a comment -->`

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Which attribute makes a form field mandatory?

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`required.`

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**Which element supplies
`alt` ernative media resources?**

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`<source>` inside `<picture>`, `<audio>`, or `<video>`.

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What are the four parts of the CSS box model?

CSS and JavaScript

Content, padding, border, and margin.

content | padding | border | margin

CSS and JavaScript

How do content-box and border-box differ?

CSS and JavaScript

content-box applies width to content only; border-box includes padding and border within the declared width.

```
* { box-sizing: border-box; }
```

CSS and JavaScript

What is Flexbox?

CSS and JavaScript

A one-dimensional CSS layout system for arranging items along main and cross axes.

CSS and JavaScript

Main axis versus cross axis?

CSS and JavaScript

The main axis follows `flex-direction`; the cross axis runs perpendicular to it.

```
row:    main ->  
        cross ^
```

CSS and JavaScript

What is a media query?

CSS and JavaScript

A CSS rule that applies styles only when conditions, such as viewport width, are met.

```
@media (max-width: 768px) { ... }
```

CSS and JavaScript

What is responsive design?

CSS and JavaScript

Designing layouts that adapt to different screens and viewport sizes.

CSS and JavaScript

What does `map` return?

CSS and JavaScript

A new array of the same length, with every item transformed.

```
items.map(item => item.name)
```

CSS and JavaScript

What does `filter` return?

CSS and JavaScript

A new array containing only the items that pass a test.

```
items.filter(item => item.active)
```

CSS and JavaScript

What is mutation?

CSS and JavaScript

Directly changing an existing array, object, or state value.

CSS and JavaScript

What is an immutable update?

CSS and JavaScript

Creating a new array or object instead of changing the original.

```
setItems([...items, newItem])
```

CSS and JavaScript

What is a Promise?

CSS and JavaScript

An object representing the eventual success or failure of an asynchronous operation.

CSS and JavaScript

What does `fetch` return?

CSS and JavaScript

A Promise that resolves to a Response object.

```
const response = await fetch(url)
```

CSS and JavaScript

Why check `response.ok`?

CSS and JavaScript

`fetch` usually resolves even for HTTP errors such as 404 or 500; `response.ok` identifies success.

```
if (!response.ok) throw new Error("Request failed")
```

CSS and JavaScript

What is a shallow copy?

CSS and JavaScript

A new outer array or object whose nested objects may still share references with the original.

CSS and JavaScript

What is a React component?

React, Next.js, and Debugging

A reusable piece of an interface, usually a function that returns UI.

React, Next.js, and Debugging

Props versus state?

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Props are read-only inputs from a parent; state is component-owned data that can change and trigger a re-render.

React, Next.js, and Debugging

When should you use a functional state update?

React, Next.js, and Debugging

When the next state depends on the previous state, such as
`setCount(prev => prev + 1)`.

```
setCount(prev => prev + 1)
```

React, Next.js, and Debugging

What is derived state?

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A value calculated from existing props or state instead of stored separately.

React, Next.js, and Debugging

Why do React lists need keys?

React, Next.js, and Debugging

Stable, unique keys let React match items correctly between renders.

```
items.map(item => <Row key={item.id} />)
```

React, Next.js, and Debugging

What is reconciliation?

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React's process for comparing render results and efficiently updating the DOM.

React, Next.js, and Debugging

What is unidirectional data flow?

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Data moves down through props; requests for change move up through callbacks.

React, Next.js, and Debugging

What is a stale state bug?

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Code uses an outdated state value; functional updates often prevent it.

React, Next.js, and Debugging

What is a Server Component in Next.js?

React, Next.js, and Debugging

A component rendered on the server; it is the App Router default and cannot use `use client`-only hooks or browser APIs.

React, Next.js, and Debugging

What is a Client Component?

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A component that can use state, effects, event handlers, and browser APIs; its file starts with `'use client'`.

```
"use client"
import { useState } from "react"
```

React, Next.js, and Debugging

What is hydration?

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Attaching client-side React behavior to HTML that was rendered earlier.

React, Next.js, and Debugging

What are DevTools used for?

React, Next.js, and Debugging

Inspecting HTML, CSS, JavaScript errors, computed styles, and network requests.

React, Next.js, and Debugging

What should you inspect for a failed network request?

React, Next.js, and Debugging

Request URL, HTTP method, status code, payload, and response body.

React, Next.js, and Debugging

What is the first step before asking AI to fix a bug?

React, Next.js, and Debugging

Make an independent diagnosis and capture the exact error, expected behavior, and smallest relevant code snippet.

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